

CHAPTER 1

INTRODUCTION

This chapter, divided into thirteen sections, introduces the research problem and situates it within the broader evolution of residential energy management and smart home communication security. Section 1.1 traces the transition from centralized Smart Grid infrastructure to Machine-to-Machine (M2M) enabled smart home networks and classifies domestic communication protocols across the OSI model. Section 1.2 presents a detailed technical overview of the ECHONET Lite protocol, its frame structure, and its native security posture. Section 1.3 analyzes the cybersecurity threat landscape confronting smart home networks and taxonomizes the principal attack vectors relevant to this study. Sections 1.4 and 1.5 articulate the research motivation and problem statement, respectively, while Section 1.5 additionally derives a set of formal research questions. Section 1.6 enumerates the research objectives, and Section 1.7 defines the scope, boundaries, and underlying assumptions of the study. Section 1.8 discusses the theoretical and practical contributions of this thesis, and Section 1.9 outlines the five-phase research roadmap adopted to achieve the stated objectives. Section 1.10 presents the organization of the remaining chapters, Section 1.11 discusses the academic and industrial significance of the study, Section 1.12 defines key terms used throughout the thesis, and Section 1.13 closes the chapter with a concise summary.

1.1 Overview

The global energy ecosystem is undergoing a fundamental structural transition toward the Smart Grid era and next-generation energy management frameworks. Driven by national decarbonization goals, fluctuating energy costs, and the need for peak-load balancing, traditional centralized generation networks are being replaced by automated Demand-Side Management (DSM) architectures. Within this ecosystem, Home Energy Management Systems (HEMS) serve as the primary coordination hubs connecting the utility grid to the residential domain. HEMS platforms dynamically optimize household

power consumption by interacting directly with distributed energy resources (DERs) such as solar inverters, electric vehicle (EV) chargers, battery storage units, and major residential appliances.

Historically, electricity distribution followed a strictly unidirectional model: bulk power was generated at centralized plants, stepped up for transmission, and delivered to passive residential consumers with no feedback path other than periodic metering. The Smart Grid paradigm inverts this relationship by embedding bidirectional communication, distributed sensing, and automated control throughout the distribution network, allowing utilities and consumers to jointly participate in load balancing, dynamic pricing response, and renewable energy integration. As distributed generation and storage assets have proliferated at the household level, the locus of intelligent control has progressively shifted from the utility substation to the consumer premises, giving rise to HEMS as the residential counterpart of utility-scale Energy Management Systems (EMS). This evolutionary trajectory, from centralized generation to distributed, consumer-centric orchestration, is the structural backdrop against which the security of residential M2M messaging must be understood.

The operational success of HEMS relies on the rapid proliferation of interconnected Consumer Internet of Things (IoT) nodes and domestic appliance infrastructure. Residential devices are no longer passive electrical loads; they operate as autonomous, energy-aware nodes capable of real-time telemetry reporting and automated load shedding. Machine-to-Machine (M2M) communication protocols form the foundational layer of this residential ecosystem, enabling seamless interoperability between heterogeneous edge appliances and central home gateways over local IP networks. Unlike enterprise or industrial M2M deployments, which typically operate within professionally managed and monitored network perimeters, domestic M2M ecosystems are provisioned, configured, and maintained largely by non-expert end users, over consumer-grade routers, and alongside untrusted general-purpose devices sharing the same local area network. This distinction is significant from a security standpoint: the same prop-

erties that make smart home M2M networks convenient and low-cost, plug-and-play discovery, minimal configuration overhead, and lightweight messaging, also reduce the opportunity for centralized security governance that is typical of enterprise or industrial control networks.

However, the hyper-connectivity of domestic M2M networks significantly expands the residential attack surface. Compromised local networks or rogue edge nodes introduce severe operational threats, as malicious command injection can disrupt device operations or cause artificial power spikes across local distribution networks. Protecting application-layer messaging within smart home environments is therefore essential to preserving consumer privacy, physical asset integrity, and grid reliability. Because HEMS gateways aggregate telemetry from, and issue control commands to, multiple appliances simultaneously, a single compromised message exchange can cascade into coordinated, multi-device disruption, an effect with no direct analogue in traditional single-appliance fault scenarios.

The urgency of securing residential M2M messaging is not speculative; it is grounded in a well-documented and rapidly worsening global cyber-threat trajectory. Cybersecurity Ventures estimated that worldwide cybercrime losses grew from approximately USD 3 trillion in 2015 to USD 6 trillion in 2021, and forecast that damages would reach USD 10.5 trillion annually by 2025, a growth rate of roughly 15 percent per year that would make cybercrime, if measured as an economy, larger than that of most nations (Morgan, 2020). Table 1.0 summarizes this trajectory to illustrate the scale and pace of the threat that motivates protective research at every layer of connected infrastructure, including the residential domain addressed by this thesis.

Table 1.0 Growth Trajectory of Global Cybercrime Costs
Year Estimated Global Cybercrime Cost (USD)
2015 3 trillion 2021 6 trillion 2025 (projected) 10.5 trillion
Source: Adapted from Cybersecurity Ventures (Morgan, 2020).

Connected consumer and IoT devices, the same class of edge nodes that HEMS platforms rely upon, have historically been at the center of some of the most damaging cyber-incidents on record, rather than a marginal or theoretical risk category. The Mirai botnet, first identified in August 2016, propagated by scanning the internet for IoT devices, including home routers, IP cameras, and other consumer-grade appliances, still configured with factory-default credentials, and enlisting them into a botnet capable of launching massive distributed denial-of-service (DDoS) attacks (Antonakakis et al., 2017). In September 2016, a Mirai-driven attack struck the Krebs on Security website at a then-record 620 Gbit/s, and the following month a coordinated Mirai attack against the Dyn DNS provider rendered major internet services, including Twitter, Netflix, Reddit, and GitHub, inaccessible across large parts of the United States and Europe (Antonakakis et al., 2017; Fruhlinger, 2018). Later that year, a related Mirai variant crashed approximately 900,000 consumer routers belonging to Deutsche Telekom customers through a failed exploitation attempt, causing widespread residential internet outages (Antonakakis et al., 2017). Rutgers University, targeted by Mirai-based DDoS attacks between 2014 and 2016, reportedly spent over USD 300,000 on incident consultation and increased its cybersecurity budget by a further USD 1 million in direct response, costs the university itself cited among its reasons for raising student tuition and fees the following year (Antonakakis et al., 2017). Critically, the threat posed by compromised consumer devices has not receded with time; it has intensified. A resurgent Mirai-based botnet, publicly identified in 2024 and comprising an estimated 300,000 or more compromised consumer devices, has continued to set new records for attack scale, including a 31.4 Tbps DDoS event recorded against Cloudflare infrastructure, underscoring that the aggregation of insecure, network-connected consumer devices remains an active and escalating global threat rather than a historical footnote (Antonakakis et al., 2017). These documented incidents demonstrate, with concrete financial and operational evidence, what can occur when large populations of under-secured, network-connected consumer devices are left without adequate origin authentication: services fail at national scale, institutions absorb unplanned costs in the hundreds of

thousands to millions of dollars, and the compromised devices themselves become instruments of harm against third parties. HEMS-integrated appliances communicating over ECHONET Lite occupy precisely this same at-risk device category, sharing direct network reachability, constrained processing resources, and, in the absence of the mechanism proposed in this thesis, an identical lack of native origin authentication. The difference is one of consequence rather than likelihood: where a compromised router or camera primarily threatens network availability, a compromised HEMS appliance can directly manipulate physical energy loads, elevating the stakes from service disruption to potential physical and grid-level harm. This precedent of demonstrated, quantifiable loss from unauthenticated consumer devices is precisely what justifies the application-layer authentication mechanism developed in this thesis.

1.2 ECHONET Lite Protocol: An Overview

Established by the ECHONET Consortium and recognized under ISO/IEC 14543-4-3, the ECHONET Lite protocol specification defines standard communication for smart home appliances and HEMS devices. Enforced across international markets and widely adopted in residential deployments, ECHONET Lite normalizes multi-vendor interoperability across more than 100 device classes, including HVAC units, smart meters, heat pump water heaters, and photovoltaic arrays.

ECHONET Lite occupies a distinctive position among smart home messaging standards because it was developed specifically with energy management semantics at its core, rather than as a general-purpose IoT messaging layer later adapted for energy use cases. Its object-oriented device model, in which every appliance class exposes a standardized set of properties (e.g., operational status, power consumption, set-point temperature), allows HEMS controllers to interoperate with appliances from different manufacturers without bespoke integration work. This device-class standardization has made ECHONET Lite the de facto messaging backbone for HEMS certification programs in several national smart grid initiatives, embedding it deeply into residential energy policy infrastructure rather than treating it as an optional convenience layer. The

protocol’s continued longevity, despite the emergence of newer general-purpose smart home standards such as Matter, is attributable to this entrenched role in energy-sector certification and the substantial installed base of already-deployed HEMS-compliant appliances.

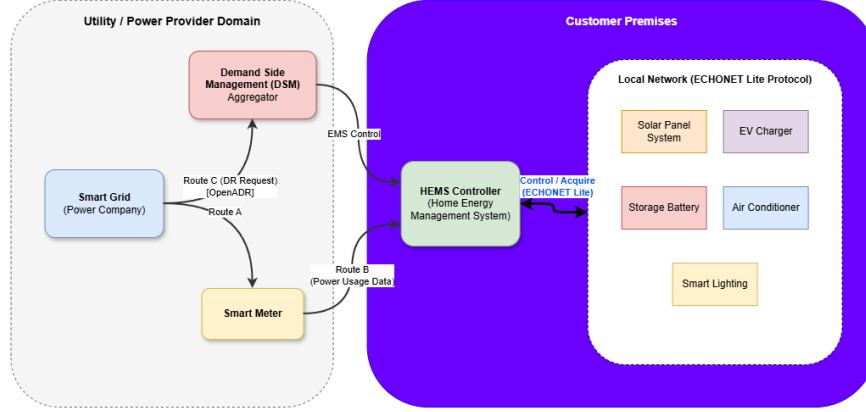


Figure 1.1: Echonet Lite Architecture.

ECHONET Lite uses an object-property-action model to govern device behavior. Devices send read, write, or notification requests (Get, Set, SNA) over unicast or multi-cast IP channels. The ESV command code field, in particular, determines the semantic intent of a frame (e.g., ‘0x62’ for a property read request, ‘0x61’ for a property write request, ‘0x73’ for an unsolicited property notification), and it is precisely this field, together with the EDT payload it governs, that an attacker targets when attempting unauthorized command injection or replay. Because the SEOJ/DEOJ addressing scheme identifies devices purely by class and instance codes rather than by any cryptographically verifiable identity, any node capable of observing or injecting traffic on the local network segment can construct a syntactically valid ECHONET Lite frame addressed to any other device on that segment.

Despite its widespread structural adoption, standard ECHONET Lite transmits plaintext binary frames without mandatory application-layer cryptographic encipherment or dynamic origin authentication. Designed originally for physically closed, trusted home networks, native ECHONET Lite frames lack fields for data confidentiality, frame integrity, or sender identity verification. Wrapping ECHONET Lite in

heavy lower-layer transport security protocols (such as DTLS or TLS) often introduces excessive processing overhead, memory footprint expansion, and latency spikes on resource-constrained 8-bit or 16-bit appliance microcontrollers. In practice, this leaves implementers with an unappealing binary choice: ship devices with no meaningful application-layer protection, or absorb the disproportionate computational cost of a full transport-security handshake on hardware that was never provisioned with that overhead in mind. This unresolved tension between security completeness and embedded resource constraints is the central technical gap that motivates the present research.

1.3 Smart-Home Networks and Cybersecurity Threats

The increasing adoption of Internet of Things (IoT) technologies has transformed conventional residential environments into interconnected smart-home ecosystems in which sensing, communication, computation, and physical control are increasingly integrated. Unlike traditional household appliances that operated independently, smart-home devices can exchange information with one another, communicate with home gateways and cloud services, and receive commands from users through local or remote interfaces. This connectivity provides significant benefits in terms of convenience, energy management, automation, and remote monitoring; however, it also introduces cybersecurity risks because devices that were previously isolated from communication networks are now exposed to a wider range of digital interactions. Security and privacy have consequently become important considerations in the design and deployment of smart-home systems (Kuyucu et al., 2019; Wang et al., 2021).

The security characteristics of smart-home networks differ from those of conventional enterprise networks. Enterprise environments are generally supported by dedicated information-technology personnel, centrally managed security policies, controlled network infrastructure, and continuous monitoring. In contrast, smart-home environments commonly contain heterogeneous devices from multiple manufacturers, consumer-grade network equipment, mobile applications, cloud services, and general-purpose personal devices operating within the same residential environment. Many of

these devices are designed with limited computational, storage, and energy resources, which can constrain the implementation of sophisticated security mechanisms. Furthermore, residential users may have limited technical knowledge and may not possess the expertise required to configure and continuously maintain the security of every connected device. Studies of smart-home users have consequently identified a gap between users' awareness of security and privacy risks and their ability to implement effective mitigation measures (Haney et al., 2020).

The increasing number of interconnected devices consequently expands the attack surface of the residential network. A compromise of one device may provide an attacker with an opportunity to interact with other devices or services within the same environment, particularly when network segmentation and access controls are insufficient. This concern extends beyond conventional information security because smart-home devices frequently interact with physical systems. For example, a compromised thermostat, lighting controller, door lock, surveillance camera, energy-management device, or appliance may allow an attacker to alter physical conditions within the residence. In energy-oriented smart homes, the consequences can extend further because Home Energy Management Systems (HEMS) can coordinate multiple appliances and distributed energy resources. Consequently, unauthorized manipulation of a single communication exchange may affect the operation of several interconnected devices simultaneously. The cyber-physical nature of smart-home environments therefore creates security consequences that may involve not only information and privacy but also physical safety, household operations, and energy management (Lemay et al., 2018).

Cybersecurity threats against smart-home networks can be considered in relation to the fundamental security requirements of confidentiality, integrity, and availability. Confidentiality concerns the protection of information from unauthorized disclosure, which is particularly important in residential environments because connected devices can generate information capable of revealing household activities, occupancy patterns, energy consumption, and user behaviour. Integrity concerns ensuring that information

and commands are not modified by unauthorized parties. This is particularly significant for smart-home control systems because an altered command may cause an appliance to perform an operation that was not requested by the legitimate user. Availability concerns ensuring that legitimate users and devices can access required services when needed. A denial-of-service condition affecting a gateway, controller, or communication channel may therefore prevent legitimate control and monitoring operations. Research on smart-home cyber-physical threats similarly demonstrates that attacks against connected residential systems can produce consequences extending from information disclosure to unauthorized control and physical effects (Lemay et al., 2018; Buil-Gil et al., 2022).

The practical significance of these vulnerabilities has been demonstrated by attacks against consumer IoT devices. One of the most prominent examples is the Mirai botnet, which demonstrated how large populations of inadequately secured IoT devices could be compromised and converted into resources for distributed denial-of-service (DDoS) attacks. Antonakakis et al. (2017) reported that the Mirai botnet reached a peak of approximately 600,000 infections and demonstrated how relatively simple attacks against embedded and consumer IoT devices could be aggregated to generate attacks against major Internet services. The affected device categories included routers, cameras, digital video recorders, and other embedded devices commonly found within consumer environments.

The Mirai incident illustrates an important characteristic of smart-home cybersecurity: an individual device may represent only a small computational resource, yet the collective compromise of a large number of such devices can produce significant consequences. Consequently, the security of a smart-home ecosystem cannot be evaluated solely according to the importance of individual appliances. Instead, the interactions between devices, gateways, communication protocols, cloud services, and users must also be considered. This interconnectedness creates opportunities for compromised devices to act as entry points, propagation points, or attack resources within a larger network. NIST's assessment of consumer home IoT products similarly

identified security concerns across multiple categories of devices, including cameras, doorbells, thermostats, smart plugs, televisions, and lighting products, highlighting the heterogeneous security characteristics of consumer IoT environments.

In addition to device compromise, weaknesses in communication and authentication mechanisms represent important attack vectors in smart-home environments. A communication protocol may provide interoperability and efficient message exchange without necessarily providing sufficient mechanisms for establishing the identity of the communicating parties or protecting the integrity and freshness of exchanged messages. This distinction is particularly important for residential control networks because successful delivery of a syntactically valid command does not necessarily demonstrate that the command originated from an authorized entity. Research on smart-home authentication has therefore emphasized the difficulty of designing authentication mechanisms that provide adequate security while remaining suitable for resource-constrained IoT devices (AlJanah et al., 2021).

Among the threats relevant to communication between smart-home devices are replay attacks, command-injection attacks, and denial-of-service attacks. A replay attack occurs when an adversary captures a previously valid message and retransmits it at a later time. If a receiving device does not verify message freshness, a previously legitimate command may be accepted as a new request. Research investigating smart-home environments has demonstrated that replay attacks are not merely theoretical; systematic testing of commercial IoT devices has identified devices that accept replayed communication under certain conditions. Such findings demonstrate the importance of incorporating freshness validation into communication-security mechanisms where commands can affect device behaviour.

An injection or message-manipulation attack represents another significant threat. In this scenario, an attacker attempts to introduce a fabricated command or modify an existing message so that the receiving device performs an unauthorized operation. The impact of such an attack depends on the functionality controlled by the affected message.

For an ordinary sensor, manipulation may result primarily in inaccurate monitoring information. For an actuator or energy-management appliance, however, unauthorized modification may directly alter the physical state of the device. Smart-home threat taxonomies have consequently identified unauthorized control and manipulation of cyber-physical systems as important attack consequences because digital attacks can ultimately affect physical processes within the home (Lemay et al., 2018).

Denial-of-service attacks constitute a further threat to smart-home communication. IoT devices are frequently constrained in terms of processing power, memory, network bandwidth, and energy availability. An attacker who generates excessive requests or malformed messages may therefore consume resources required for legitimate communication. At the network level, compromised IoT devices may also participate in distributed attacks against external targets, as demonstrated by Mirai. At the individual smart-home level, disruption of communication can prevent users or HEMS controllers from monitoring and controlling connected appliances. Therefore, availability must be considered alongside authentication and integrity when evaluating the security of residential communication networks.

The security problem is further complicated by the role of the residential router and home gateway. These devices frequently provide the communication boundary between internal smart-home devices and external networks, while also providing connectivity for conventional computers, smartphones, and other personal devices. A compromise of the router can therefore affect multiple devices simultaneously. NIST has specifically highlighted the importance of securing consumer-grade routers because compromised routers can be used to compromise or attack other devices and networks connected through them. This makes the router and gateway important security components within the smart-home architecture rather than merely communication infrastructure.

Another important characteristic of the smart-home threat landscape is the limited ability of residential users to manage security across heterogeneous devices. Smart-home users may purchase devices independently from different manufacturers, each

potentially implementing different authentication methods, update mechanisms, communication protocols, and security configurations. This creates a fragmented security environment in which the overall security of the home may depend on the weakest component. NIST’s consumer IoT baseline consequently emphasizes security capabilities that should apply across the product lifecycle rather than relying entirely on end-user configuration. Similarly, empirical studies of smart-home users have shown that consumers frequently prioritize functionality and convenience while possessing limited options or knowledge for implementing sophisticated security controls (Megaw et al., 2022; Haney et al., 2020).

These challenges become particularly relevant in smart homes that employ HEMS for automated energy management. HEMS acts as a coordination point between appliances, distributed energy resources, and users, and may therefore exchange both monitoring information and control commands with multiple devices. The communication protocol used between these components becomes an important security boundary. If messages are accepted solely because they originate from a reachable node on the local network, network access may effectively become equivalent to communication authorization. Such an assumption is increasingly inappropriate as smart-home networks incorporate remote access, cloud connectivity, mobile applications, and heterogeneous third-party devices. A secure communication architecture must therefore distinguish between network reachability and authorization to issue or modify control messages.

For the ECHONET Lite-based environment considered in this study, this distinction is particularly important. ECHONET Lite provides an application-level communication framework for exchanging information between HEMS and smart-home appliances, but the existence of a standardized message format does not inherently establish the legitimacy of the sender. As discussed in Section 1.2, ECHONET Lite was designed for communication within residential environments and its standard message structure does not inherently provide the application-layer security properties required to establish message origin, verify message integrity, and prevent the unauthorized reuse of

previously valid commands.

Accordingly, the cybersecurity problem addressed by this research is not the complete security of the smart-home ecosystem, but the security of messages exchanged at the application layer. The relevant threat model focuses on an adversary capable of observing, injecting, modifying, or replaying communication within the network path. Under such conditions, message authentication is required to establish whether a message originated from an authorized communicating entity, integrity protection is required to detect unauthorized modification, and freshness validation is required to prevent previously valid messages from being accepted repeatedly. These requirements provide the security foundation for the mechanism investigated in this research.

The increasing connectivity of residential devices therefore changes the security assumptions under which smart-home communication protocols were originally deployed. Smart-home networks can no longer be considered inherently trustworthy merely because communication occurs within a residential environment. The coexistence of consumer IoT devices, remote access mechanisms, heterogeneous vendors, resource-constrained appliances, and Internet-connected gateways creates a broader attack surface that requires security to be incorporated directly into communication mechanisms. For the HEMS environment examined in this study, ensuring message authentication, integrity, and freshness is consequently an important step toward preventing unauthorized control of connected appliances while maintaining the efficiency required by resource-constrained residential devices.

1.4 Research Motivation

The motivation for this research is the gap between the lightweight appliance communication model of ECHONET Lite and the security assurance required when that communication is exposed to an untrusted or remotely reachable network. An application-layer mechanism can make the security decision at the point where the ECHONET Lite message is parsed and interpreted.

The proposed direction is a frame-level extension that authenticates relevant message data, detects modification, and rejects commands that fail freshness or replay validation. Its usefulness must be assessed not only by security behaviour but also by the communication and processing overhead imposed on constrained appliance nodes.

1.5 Problem Statement

This study addresses the following problem: standard ECHONET Lite communication provides an application-level data model for home appliances, but the standard frame does not by itself provide the message authentication, integrity verification, and anti-replay controls required for communication across a potentially untrusted network (**echonet_lite_part2**). Without a mechanism that combines origin verification, message integrity, and freshness validation, an attacker who gains access to the communication path may attempt unauthorized command injection or replay.

1.6 Research Objectives

The objectives of this research are to achieve the following:

1. To propose a frame-level security mechanism for ECHONET Lite that provides message authentication and data integrity without changing the protocol's core object and property model.
2. To propose an anti-replay and temporal-validation mechanism within the security extension to reduce the risk of unauthorized command injection and duplicated valid messages.
3. To evaluate the proposed mechanism in terms of security behavior and performance overhead, including its effect on latency and operational efficiency relative to standard ECHONET Lite communication.

1.7 Scope and Limitations

The study is limited to application-layer protection for ECHONET Lite messages. It focuses on message authentication, data integrity, freshness or replay validation, and the processing overhead associated with the proposed extension. It does not claim to solve every security problem in a smart grid or smart home. Physical-layer attacks, radio interference, appliance hardware tampering, general-purpose public-key infrastructure, and the security of unrelated smart-home protocols are outside the primary scope unless required for comparison.

The evaluation results will be bounded by the selected simulation, emulation, or hardware environment. Results obtained from that environment should therefore be interpreted as evidence about the tested implementation and workload, rather than as universal performance values for every ECHONET Lite appliance.

1.8 Research Contribution

The intended contribution is an application-layer security extension for ECHONET Lite that connects protocol-level message processing with authentication, integrity, and replay validation requirements. A second contribution is an implementation and evaluation procedure that makes the overhead of the extension observable. Contribution claims will be finalized only after the design and evaluation results have been reported; design features or performance values are not presented here as empirical findings.

1.9 Research Steps

The research follows five stages. First, the ECHONET Lite message model and relevant threats are analysed to derive security and performance requirements. Second, the authentication and freshness mechanism is designed. Third, the mechanism is implemented and integrated with a standard ECHONET Lite communication path. Fourth, legitimate and adversarial message exchanges are tested in the selected evaluation environment. Finally, the results are analysed using the defined security and

performance measures.

1.10 Thesis Organization

Chapter ?? reviews ECHONET Lite, smart-home communication, and relevant security mechanisms. Chapter ?? presents the requirements and design of the proposed extension. Chapter ?? describes its implementation and test environment. The discussion chapter interprets the evaluation results and considers limitations, while the conclusion summarizes the findings and identifies possible future work.